

AI Smart Attendance System

Day-1 Deliverable: Evaluation & Selection of Face Detection/Recognition Models

Author: Priti Renuke

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Project: Smart Attendance System

1. Executive Summary

For an automated AI Smart Attendance System, choosing the right combination of face detection and recognition algorithms is critical to achieving high accuracy under varying lighting conditions, camera angles, and student density. This deliverable presents a comprehensive evaluation of five widely-used framework models and proposes a hybrid operational pipeline.

2. Comparative Framework Analysis

FRAMEWORK	TYPE	KEY ADVANTAGES	LIMITATIONS	SUITABILITY
OpenCV Haar Cascades	Detection	Ultra-lightweight, extremely fast on low-end CPUs, minimal resource consumption.	High false positive rate, highly sensitive to lighting changes and side profiles.	Low
MediaPipe	Detection & Alignment	Blazing fast real-time CPU performance, highly accurate bounding box & 3D landmark tracking.	Does not generate facial embeddings or perform identification/recognition.	High (Detect)
MTCNN	Detection & Alignment	Multi-stage cascade network offering high precision and built-in 5-point alignment.	Slower execution latency on dense frame streams compared to single-shot detectors.	Medium
face_recognition (dlib)	Detection & Recognition	Simple Python API, high benchmark accuracy (99.38%), fast 128-d vector embedding.	Heavy C++ dlib dependencies; limited multi-threading hardware acceleration on edge.	Medium
DeepFace	Wrapper Framework	Supports SOTA backbones (ArcFace, FaceNet, VGG-Face), handles face verification & analysis.	Higher memory footprint and initialization overhead.	High (Recog)

Evaluation Note:

The current comparison is based on framework capabilities, documented characteristics, and project suitability. Empirical accuracy, processing time, and resource-usage testing will be performed on the project dataset during the implementation and testing phase before finalizing the production configuration.

3. Recommended Operational Approach

Proposed Architecture: Two-Stage Hybrid Pipeline

To maximize system throughput and accuracy in a classroom environment, a decoupled hybrid architecture is recommended:

1. **Stage 1 — Face Detection & Alignment (MediaPipe):** Process the uploaded classroom/group photo using MediaPipe for fast face detection, landmark-based alignment, and face cropping across multiple faces in the image.
2. **Stage 2 — Feature Extraction & Verification (DeepFace / ArcFace):** Pass cropped face images into DeepFace (utilizing the ArcFace or FaceNet backend) to compute high-dimensional feature vectors and match them against the student vector database.